

Physics 1502Q

Tutorial 6.2

Midterm 1 Exam Review

There is no correlation between these practice problems and the actual exam problems.

Problem 1

A positive point charge Q is fixed on a very large horizontal frictionless tabletop. A second positive point charge q is released from rest near the stationary charge Q and is free to move. Which statement best describes the motion of q after it is released? **Justify**.

- A. Its speed will be greatest just after it is released.
- B. Its acceleration is zero just after it is released.
- C. As it moves farther and farther from Q , its acceleration will increase.
- D. As it moves farther and farther from Q , its acceleration will decrease.

Problem 1 - Solution

A positive point charge Q is fixed on a very large horizontal frictionless tabletop. A second positive point charge q is released from rest near the stationary charge and is free to move. Which statement best describes the motion of q after it is released? **Justify**.

- A. Its speed will be greatest just after it is released.
- B. Its acceleration is zero just after it is released.
- C. As it moves farther and farther from Q , its acceleration will increase.
- D. As it moves farther and farther from Q , its acceleration will decrease.

The force on the charge (and hence acceleration) will be greatest when the charges are closest. As the charges get farther apart, the force and acceleration decreases (as $1/r^2$)

Problem 2

A sphere with charge $+q$ is placed a distance d from an uncharged metal sphere. Of the four figures shown, which figure best represents the resulting charge distribution on the metal sphere? Justify.

A. Figure A

B. Figure B

C. Figure C

D. Figure D

E. None of these

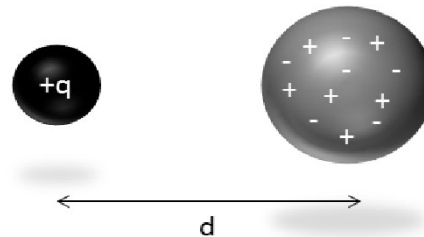


Figure A

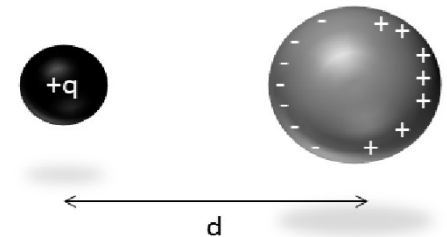


Figure B

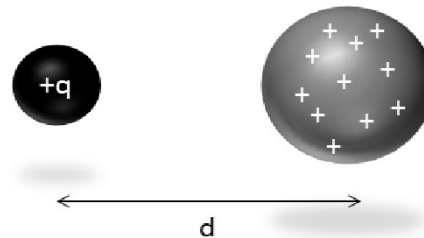


Figure C

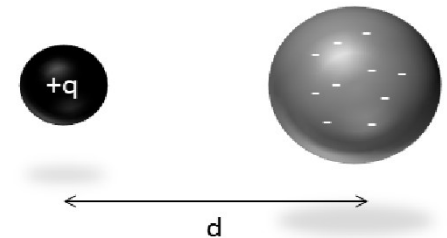


Figure D

Problem 2 - Solution

A sphere with charge $+q$ is placed a distance d from an uncharged metal sphere. Of the four figures shown, which figure best represents the resulting charge distribution on the metal sphere? **Justify.**

A. Figure A

B. Figure B

C. Figure C

D. Figure D

E. None of these

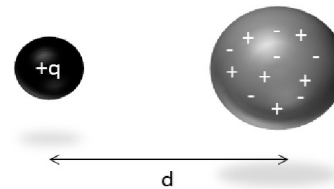


Figure A

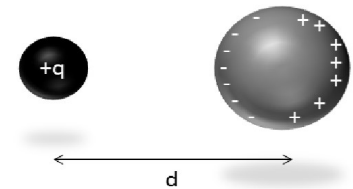


Figure B

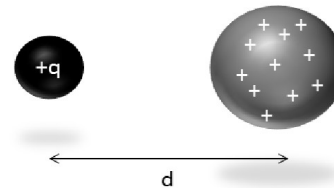


Figure C

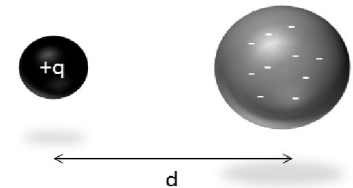
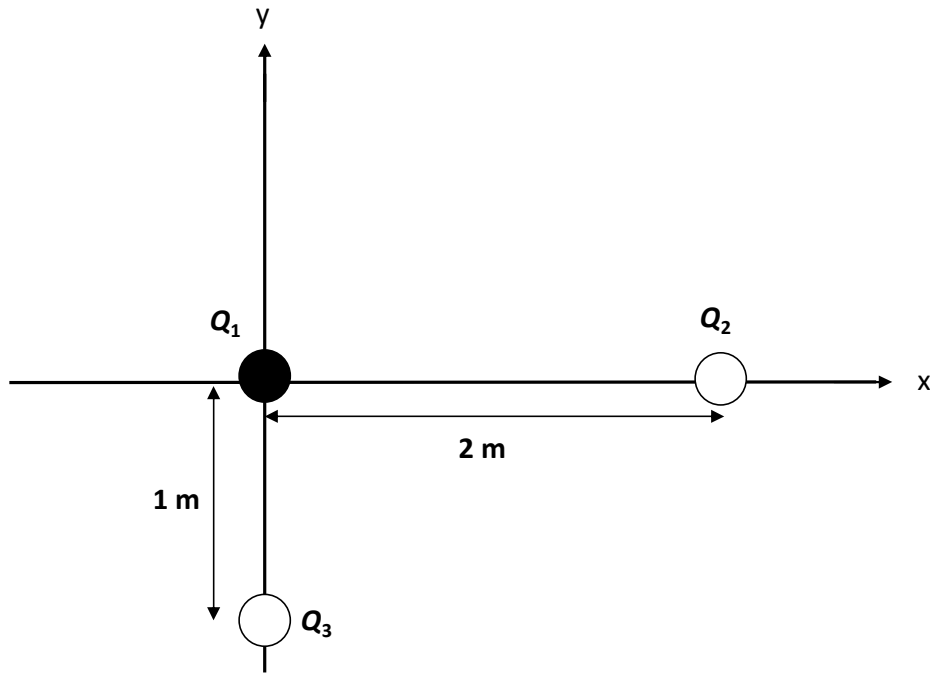


Figure D

Justification: Charges in a conductor can move freely and they will redistribute in a way that positive charges on the sphere repel the $+q$ charge and negative charges in the sphere attract $+q$.

Problem 3

Three charges are arranged as shown in the diagram below. All three charges have a value of $+1\ \mu\text{C}$. What is the net force on charge Q_2 ?



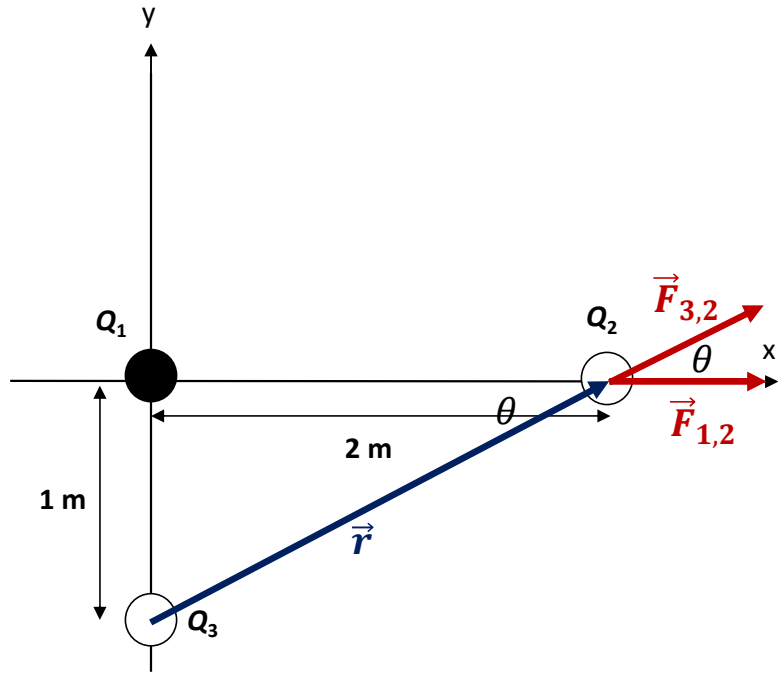
$$Q_1 = +1\ \mu\text{C}$$

$$Q_2 = +1\ \mu\text{C}$$

$$Q_3 = +1\ \mu\text{C}$$

Problem 3 - Solution

Three charges are arranged as shown in the diagram below. All three charges have a value of $+1 \mu\text{C}$. What is the net force on charge Q_2 ?



$$r = (1\text{m}^2 + 2\text{m}^2)^{\frac{1}{2}} = 2.24\text{m}$$

$$\cos\theta = \frac{2\text{m}}{r}$$

$$\sin\theta = \frac{1\text{m}}{r}$$

Net force in the x-direction:

$$\begin{aligned} F_x &= F_{1,2} + F_{3,2}\cos\theta \\ &= \frac{kQ_1Q_2}{(2\text{m})^2} + \frac{kQ_3Q_2}{(r)^2} \left(\frac{2\text{m}}{r}\right) \\ &= \frac{(9 \times 10^9 \text{Nm}^2/\text{C}^2)(1 \times 10^{-6}\text{C})^2}{(2\text{m})^2} + \frac{(9 \times 10^9 \text{Nm}^2/\text{C}^2)(1 \times 10^{-6}\text{C})^2(2\text{m})}{(2.24\text{m})^3} \\ &= 2.25 \times 10^{-3} \text{N} + 1.6 \times 10^{-3} \text{N} \\ &= 3.85 \times 10^{-3} \text{N} \end{aligned}$$

Net force in the y-direction:

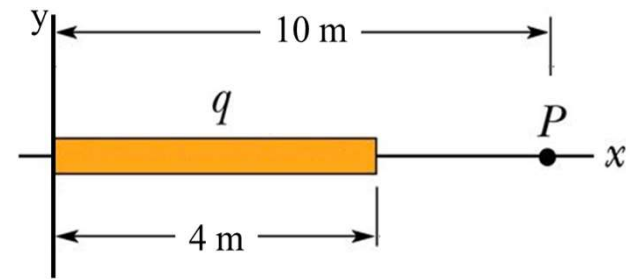
$$\begin{aligned} F_y &= F_{3,2}\sin\theta \\ &= \frac{kQ_3Q_2}{(r)^2} \left(\frac{1\text{m}}{r}\right) \\ &= \frac{(9 \times 10^9 \text{Nm}^2/\text{C}^2)(1 \times 10^{-6}\text{C})^2(1\text{m})}{(2.24\text{m})^3} \\ &= 8.0 \times 10^{-4} \text{N} \end{aligned}$$

Total Net force:

$$\vec{F} = F_x\hat{i} + F_y\hat{j} = (3.85 \times 10^{-3}\hat{i} + 8.0 \times 10^{-4}\hat{j})\text{N}$$

Problem 4

A thin rod has a total charge q distributed uniformly along the x -axis from $x = 0$ to $x = 4$ m. Set up the integral that gives the x -component of the electric field at point P at a distance of 10 m from the origin on the x -axis?



Problem 4 - Solution

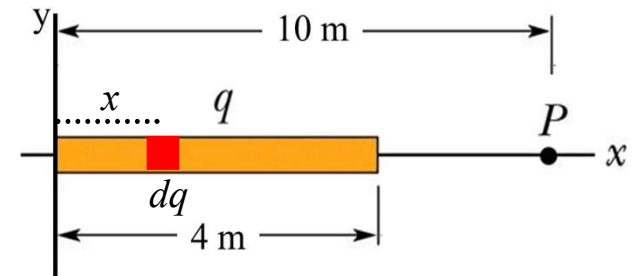
A thin rod has a total charge q distributed uniformly along the x -axis from $x = 0$ to $x = 4$ m. Set up the integral that gives the x -component of the electric field at point P at a distance of 10 m from the origin on the x -axis?

$$|\vec{E}| = \int k \frac{dq}{r^2}$$

$$\lambda = \frac{dq}{dx} \quad \longrightarrow \quad dq = \lambda dx$$

$$E_x = k\lambda \int_0^{4\text{ m}} \frac{dx}{(10\text{ m} - x)^2} \quad \text{Since } \lambda = \frac{q}{4\text{ m}}$$

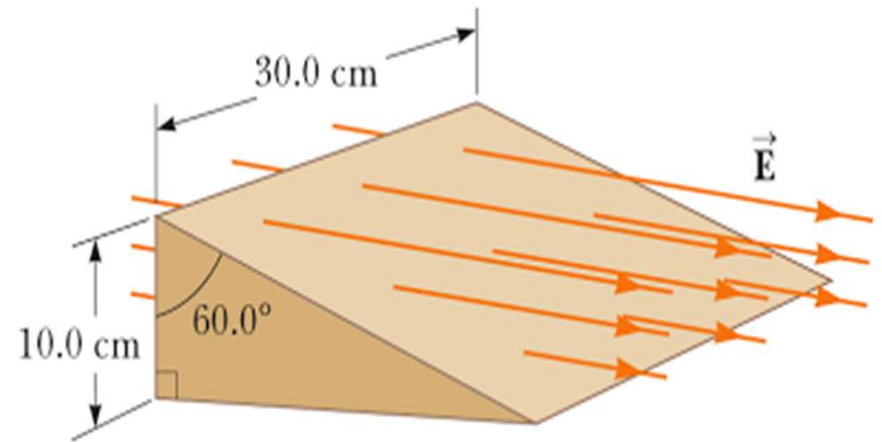
$$E_x = k \frac{q}{4\text{ m}} \int_0^{4\text{ m}} \frac{dx}{(10\text{ m} - x)^2}$$



Problem 5

Consider a closed triangular box resting within a horizontal electric field of magnitude $E = 7.80 \times 10^4 \text{ N/C}$ as shown in the figure.

Calculate the electric flux through the vertical rectangular surface.



Problem 6

What is the sign of the flux through the slanted surface? Justify.

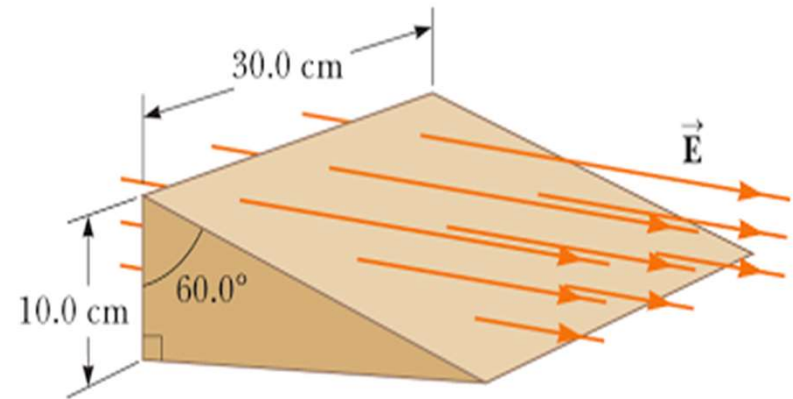
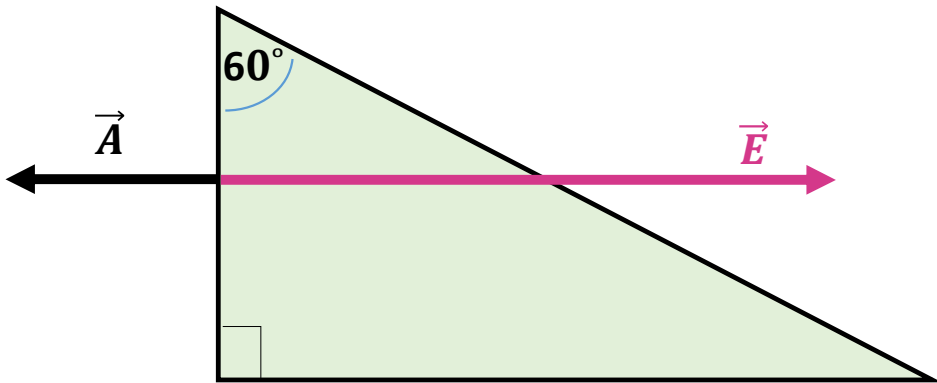
- A. $\Phi_{slanted} > 0$
- B. $\Phi_{slanted} < 0$
- C. $\Phi_{slanted} = 0$

Problem 5 - Solution

Consider a closed triangular box resting within a horizontal electric field of magnitude $E = 7.80 \times 10^4 \text{ N/C}$ as shown in the figure.

Calculate the electric flux through the vertical rectangular surface.

Cross-section of the wedge



Angle between $\vec{E}$ and $\vec{A}$ is 180° .

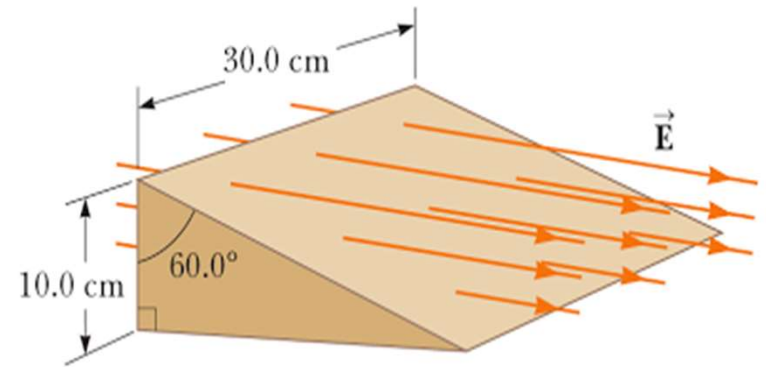
$$\begin{aligned}\Phi_E &= \vec{E} \cdot \vec{A} = E A \cos(180^\circ) \\ &= (7.8 \times 10^4 \text{ N/C})(0.3 \text{ m} \times 0.1 \text{ m}) \cos(180^\circ) \\ &= \boxed{-2340 \text{ Nm}^2/\text{C}}\end{aligned}$$

Problem 6 - Solution

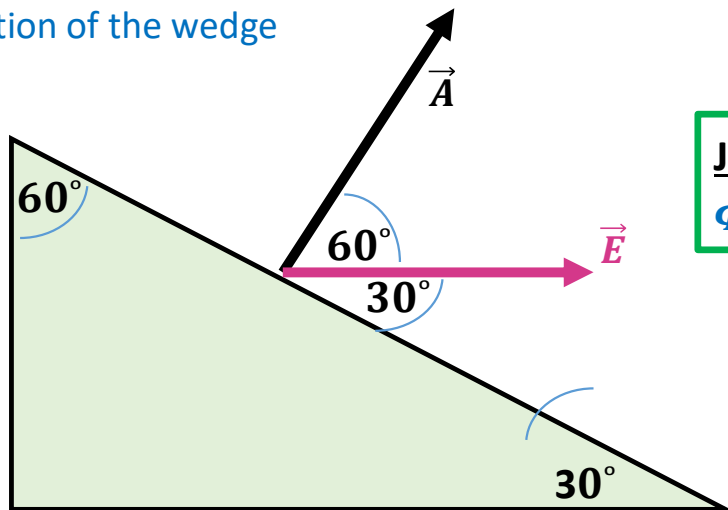
Consider a closed triangular box resting within a horizontal electric field of magnitude $E = 7.80 \times 10^4 \text{ N/C}$ as shown in the figure.

What is the sign of the flux through the slanted surface? Justify.

- A. $\Phi_{\text{slanted}} > 0$
- B. $\Phi_{\text{slanted}} < 0$
- C. $\Phi_{\text{slanted}} = 0$



Cross-section of the wedge



Justification: The angle between $\vec{E}$ and $\vec{A}$ is less than 90° thus $\Phi_{\text{slanted}} > 0$. (Here angle between $\vec{E}$ and $\vec{A}$ is 60°)

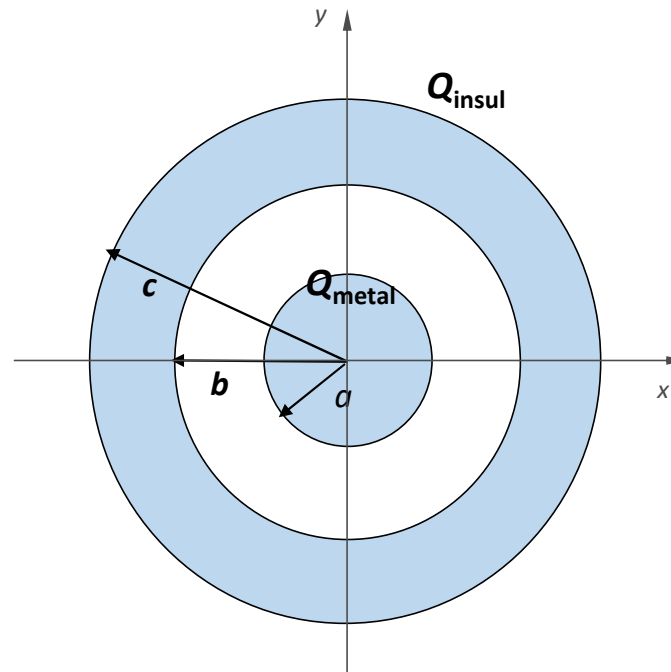
Problems 7, 8 & 9

A long metal (perfectly conducting) cylinder of radius a and length L is concentric with the z -axis. Surrounding it, and also concentric with the z axis, is a cylindrical shell made of insulating material; it is of the same length L , has inner radius b and outer radius c . (Both cylinders are so long compared with their radii that they may be considered of infinite length) A total negative charge Q_{metal} is placed on the inner metal cylinder, while a total positive charge Q_{insul} is uniformly distributed over the volume of the outer insulating cylinder. The values of all parameters are given in the figure.

7. Calculate the magnitude of the electric field E at a radius of 5 cm from the z -axis.

8. What is the surface charge density σ_{metal} on the outer surface of the metal cylinder?

9. Calculate the magnitude of the electric field E at a radius of 7 cm from the z -axis.



**Both cylinders are of
length $L = 25$ m
(into the page)**

$a = 3$ cm
$b = 6$ cm
$c = 9$ cm
$Q_{\text{metal}} = -5$ mC
$Q_{\text{insul}} = +2$ mC

Problem 7 - Solution

Calculate the magnitude of the electric field E at a radius of 5 cm from the z-axis.

- Draw the Gaussian surface at $r = 5$ cm.
- Then apply Gauss's Law.

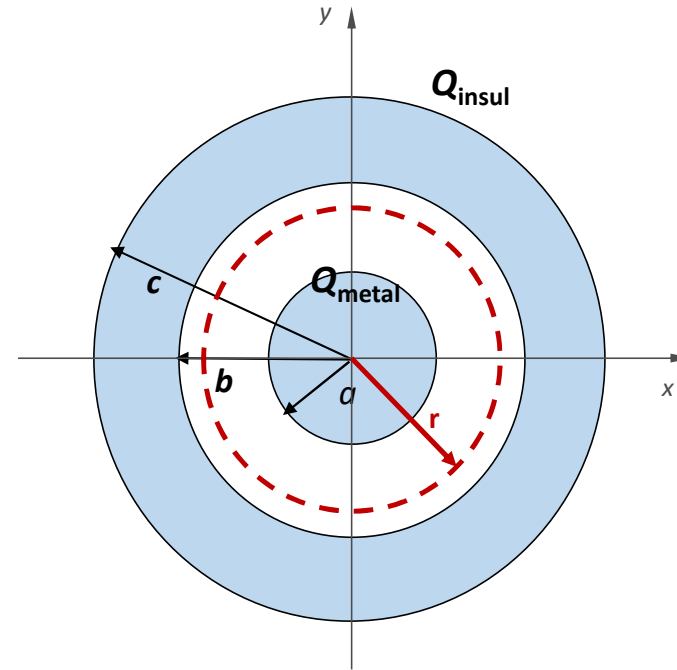
$$\text{From Gauss's Law: } \Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$$

$$\text{From cylindrical symmetry: } EA = \frac{q_{enc}}{\epsilon_0}$$

$$q_{enc} = Q_{metal} = -5 \times 10^{-6} \text{ C}$$

$$A = 2\pi rL$$

$$E = \frac{q_{enc}}{A \epsilon_0} = \frac{q_{enc}}{2\pi rL \epsilon_0} = \frac{5 \times 10^{-6} \text{ C}}{2\pi (0.05 \text{ m})(25 \text{ m})(8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2)} = 7.2 \times 10^4 \text{ N/C}$$



Both cylinders are of length $L = 25$ m (into the page)

$$\begin{aligned} a &= 3 \text{ cm} \\ b &= 6 \text{ cm} \\ c &= 9 \text{ cm} \\ Q_{metal} &= -5 \text{ mC} \\ Q_{insul} &= +2 \text{ mC} \end{aligned}$$

When calculating the magnitude of the electric field, the negative sign of Q_{metal} doesn't matter.

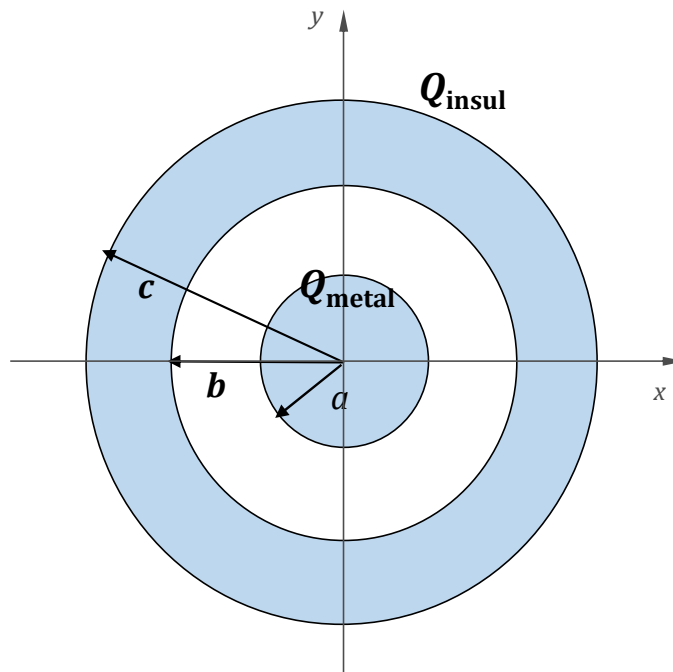
Problem 8 - Solution

What is the surface charge density σ_{metal} on the outer surface of the metal cylinder?

Surface charge density: $\sigma = \frac{Q}{A}$

$$\begin{aligned}\sigma_{metal} &= \frac{Q_{metal}}{2 \pi a L} \\ &= \frac{-5 \times 10^{-6} \text{ C}}{2 \pi (0.03 \text{ m})(25 \text{ m})}\end{aligned}$$

$$= -1.06 \times 10^{-6} \text{ C/m}^2$$



Both cylinders are of
length $L = 25 \text{ m}$
(into the page)

$$\begin{aligned}a &= 3 \text{ cm} \\ b &= 6 \text{ cm} \\ c &= 9 \text{ cm} \\ Q_{metal} &= -5 \text{ } \mu\text{C} \\ Q_{insul} &= +2 \text{ } \mu\text{C}\end{aligned}$$

Problem 9 - Solution

Calculate the magnitude of the electric field E at a radius of 7 cm from the z-axis.

- Draw the Gaussian surface at $r = 7$ cm.
- Then apply Gauss's Law.

From Gauss's Law: $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{enc}}{\epsilon_0}$

From cylindrical symmetry: $E = \frac{q_{enc}}{A\epsilon_0}$ Where $A = 2\pi rL$

What is q_{enc} here?

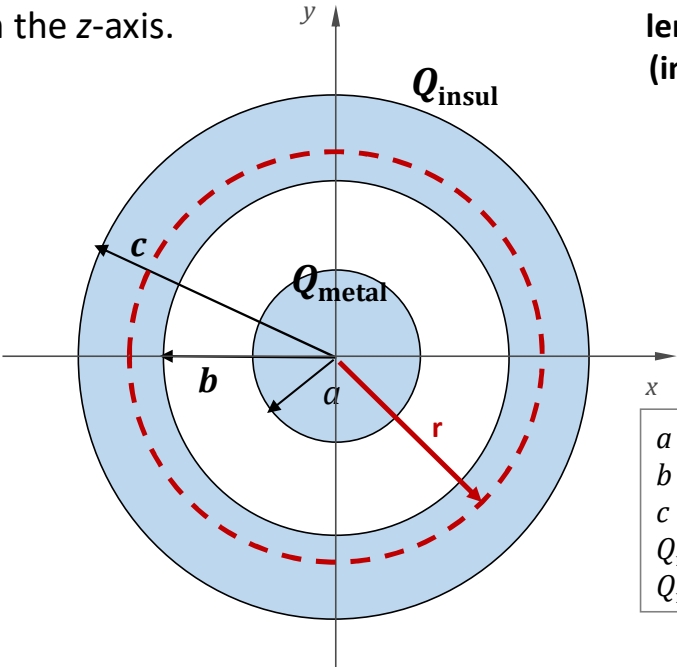
For the insulator: Volume charge density $\rho = \frac{Q_{insul}}{\pi(c^2 - b^2)L}$

$$q_{enc} = \rho (\pi(r^2 - b^2)L) + Q_{metal}$$

$$= \left(\frac{Q_{insul} (r^2 - b^2)}{c^2 - b^2} \right) + Q_{metal} = \left(\frac{(2 \times 10^{-6} C)[(0.07 m)^2 - (0.06 m)^2]}{(0.09 m)^2 - (0.06 m)^2} \right) + (-5 \times 10^{-6} C) = -4.42 \times 10^{-6} C$$

Note:
Volume of a cylinder $V = \pi r^2 L$

$$E = \frac{q_{enc}}{A \epsilon_0} = \frac{q_{enc}}{2\pi r L \epsilon_0} = \frac{4.42 \times 10^{-6} C}{2\pi (0.07 m)(25m)(8.85 \times 10^{-12} C^2/Nm^2)} = 4.54 \times 10^4 N/C$$



Both cylinders are of
length $L = 25$ m
(into the page)

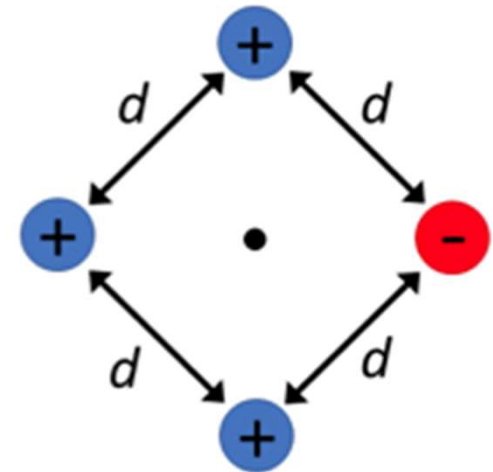
$a = 3$ cm
 $b = 6$ cm
 $c = 9$ cm
 $Q_{metal} = -5 \mu C$
 $Q_{insul} = +2 \mu C$

When calculating the magnitude of the Electric Field, the negative sign of charge doesn't matter.

Problem 10

Four point-charges are equally spaced by a distance $d = 8 \text{ cm}$ at the corners of a square as shown in the figure. Three of the charges are positive with $q = 2.9 \mu\text{C}$, while one is negative with charge $q = -2.9 \mu\text{C}$.

(a) What is the total potential energy of the configuration?



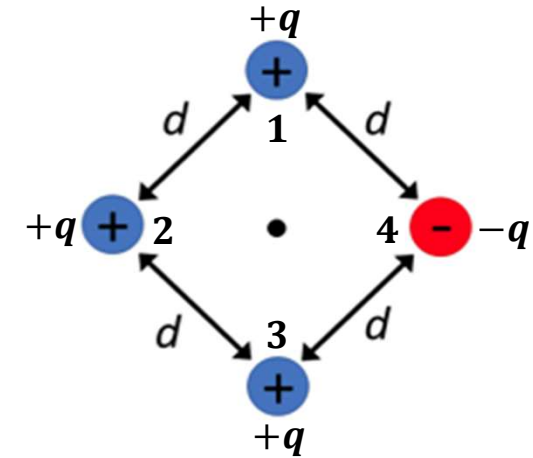
(b) What is the electric potential at center of the square?

Problem 10 - Solution

(a) What is the total potential energy of the configuration?

$$\Delta U = -W_{\text{electric}} = W_{\text{external}}$$

Bring in the charges one – by – one;



1) Change in potential energy for charge 1 $\Rightarrow U_1 = 0$

2) Change in potential energy for charge 2 $\Rightarrow U_2 = \frac{k(+q)(+q)}{d} = \frac{kq^2}{d}$

3) Change in potential energy for charge 3 $\Rightarrow U_3 = \frac{k(+q)(+q)}{d} + \frac{k(+q)(+q)}{\sqrt{2}d} = \frac{kq^2}{d} + \frac{kq^2}{\sqrt{2}d}$

4) Change in potential energy for charge 4 $\Rightarrow U_4 = \frac{k(+q)(-q)}{d} + \frac{k(+q)(-q)}{\sqrt{2}d} + \frac{k(+q)(-q)}{d} = -\frac{2kq^2}{d} - \frac{kq^2}{\sqrt{2}d}$

$$\text{Total potential energy} = U_1 + U_2 + U_3 + U_4 = 0 + \left(\frac{kq^2}{d}\right) + \left(\frac{kq^2}{d} + \frac{kq^2}{\sqrt{2}d}\right) + \left(-\frac{2kq^2}{d} - \frac{kq^2}{\sqrt{2}d}\right) = 0$$

Total energy required to bring the 4 charges one by one (ΔU) = 0

Problem 10 - Solution

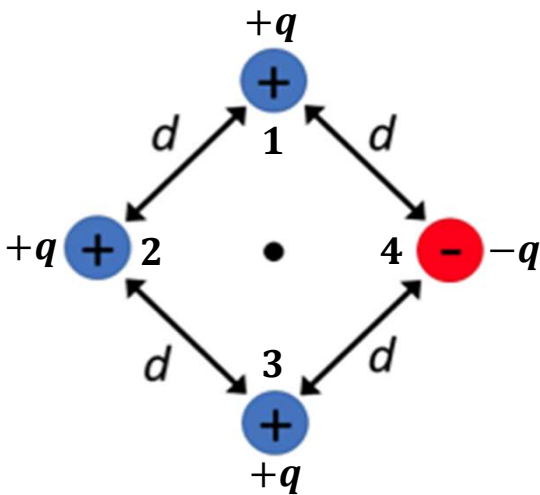
(b) What is the electric potential at center of the square?

Distance from the center of the square to each of the charges is half of the diagonal length of the square $\left(\frac{\sqrt{2} d}{2}\right)$

Electric Potential; $V = \sum_i \frac{kq_i}{r_i}$

$$= \frac{k(+q)}{\left(\frac{\sqrt{2} d}{2}\right)} + \frac{k(+q)}{\left(\frac{\sqrt{2} d}{2}\right)} + \frac{k(+q)}{\left(\frac{\sqrt{2} d}{2}\right)} + \frac{k(-q)}{\left(\frac{\sqrt{2} d}{2}\right)}$$
$$= \frac{2kq}{\left(\frac{\sqrt{2} d}{2}\right)} = \frac{2 \left(8.988 \times 10^9 \frac{Nm^2}{C^2}\right) (2.9 \times 10^{-6} C)}{\frac{\sqrt{2} \times 0.08 m}{2}}$$

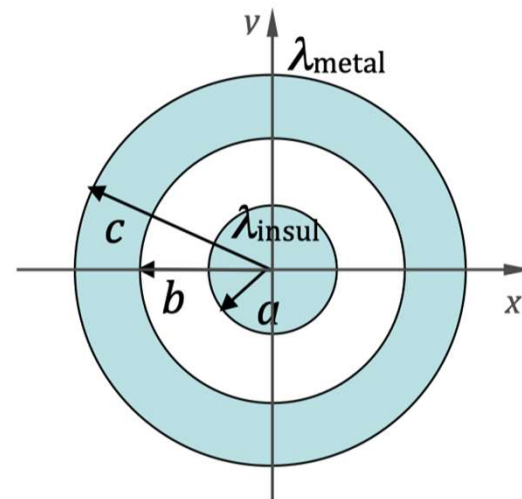
$$V = 9.215 \times 10^5 V$$



Problem 11

A long cylinder of radius a and length L is made of insulating material and is coaxial with the z -axis. Surrounding it, and also coaxial with the z -axis, is a cylindrical shell made of metal; it is of the same length L , has inner radius b and outer radius c . (Both cylinders are so long compared with their radii that they may be considered of infinite length.) Line-charge densities of λ_{insul} and λ_{metal} are placed on the inner cylinder and outer shell respectively. The values of all parameters are given in the figure.

Find the potential difference $V_a - V_c$ between the surface of the insulating cylinder ($r = a$) and the outer surface of the metal shell ($r = c$).



Both cylinders are of
length $L = 350$ m
(into page)

$$a = 5 \text{ m}$$

$$b = 11 \text{ m}$$

$$c = 17 \text{ m}$$

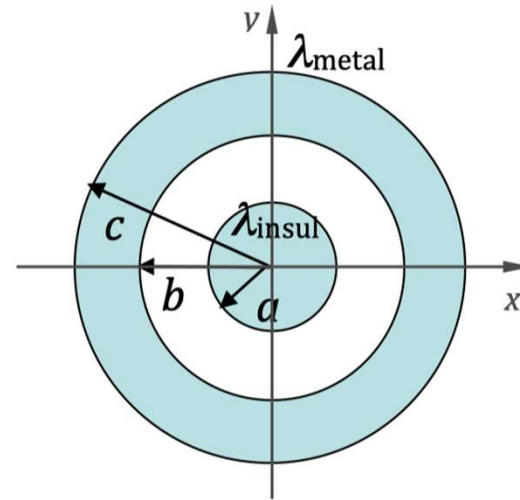
$$\lambda_{\text{insul}} = -8 \cdot 10^{-9} \text{ C/m}$$

$$\lambda_{\text{metal}} = +5 \cdot 10^{-9} \text{ C/m}$$

Problem 11 - Solution

$$\vec{E} = \frac{Q_{\text{encl.}}}{\epsilon_0 A} \hat{r} = \frac{\lambda_{\text{insul}} L}{\epsilon_0 2\pi r L} \hat{r} \quad (\text{From Gauss's law})$$

$$d\vec{s} = dr \hat{r}$$



Both cylinders are of
length $L = 350$ m
(into page)

$$a = 5 \text{ m}$$

$$b = 11 \text{ m}$$

$$c = 17 \text{ m}$$

$$\lambda_{\text{insul}} = -8 \cdot 10^{-9} \text{ C/m}$$

$$\lambda_{\text{metal}} = +5 \cdot 10^{-9} \text{ C/m}$$

$$V_a - V_c = - \int_c^a \vec{E} \cdot d\vec{s} = - \int_c^b 0 ds - \int_b^a \vec{E} \cdot d\vec{s} = - \int_b^a \frac{\lambda_{\text{insul}}}{2\pi\epsilon_0 r} dr = - \frac{\lambda_{\text{insul}}}{2\pi\epsilon_0} \ln(r) \Big|_b^a$$

$$\Rightarrow V_a - V_c = - \frac{\lambda_{\text{insul}}}{2\pi\epsilon_0} \ln\left(\frac{a}{b}\right) \simeq \boxed{-113.4 \text{ V}}$$